



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads - Executive summary

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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What happened

On 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated to carry 96 passengers, was operating service TF18. This was the last service of the day, returning from Muntari Island. On board were 38 passengers, a master, a deckhand and a kiosk attendant. Earlier that day, at the change of master at the town wharf, the outgoing master told the relieving master that the chart plotter (the electronic chart display showing the vessel's position) had frozen twice that morning and had recovered by itself both times. The fault was not entered in the defect book. The ferry departed at 17:35 local time, after sunset (17:02) and after the end of civil twilight (17:29). Under the winter timetable this evening service had run in darkness every day since 01/06/2026. The ferry entered the Weerona Channel from the north at 18:03:58 and held the charted centreline to within 3 m at 8.4 kt. At 18:04:14 the deckhand was sent from the bridge to collect fares. At 18:07:06 the chart plotter display froze, showing the vessel symbol stationary, and it did not recover. The vessel held a heading of 199 degrees, which carried it westward. At 18:09:00 it crossed the western edge of the dredged channel at 8.0 kt, with no change to the throttles. At 18:08:31 the master began looking for the No. 4 beacon and sighted it at 18:10:02. A later check found that the beacon cannot be seen from the channel over the stretch the vessel covered between those two times.

At 18:11:48 the echo sounder (depth sounder) shallow-water alarm sounded. Four seconds later both throttles were put to astern. The ferry travelled 58 m before it struck Boolarong Bank at 18:12:04 at 6.8 kt. Its trials stopping distance from 8 kt is 92 m. It came to rest within 11 m, 74 m west of the charted centreline and 54 m outside the dredged channel, heeled 4 degrees to starboard, on a falling tide. At 18:13 a pan pan (urgency) broadcast was made from the vessel, stating the decision not to attempt to refloat. Port Control called out the port workboat at 18:13:29. All passengers were in lifejackets by about 18:22, although those in the forward saloon were delayed by the location of the lifejacket locker behind the kiosk counter. The holed starboard compartment was checked from 18:16 and every ten minutes after that. The workboat came alongside at 18:41 and the last passenger was ashore at 19:26. No one was seriously injured. One passenger reported a grazed shin and two were treated at the wharf for distress. The ferry was substantially damaged: the starboard hull bottom was split and flooded to approximately 260 mm, and the starboard propeller, shaft, rudder and skeg (the fin under the hull) were damaged. The vessel refloated on the tide at 00:20 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

Why it happened

The ferry grounded because it left the dredged channel and tracked onto Boolarong Bank. It is virtually certain that this departure from the channel contributed to the grounding. The factor with the greatest influence was that the transit continued at eight knots after the chart plotter froze, rather than the vessel stopping in the channel. It is virtually certain that this occurred. It is extremely likely that it contributed both to the vessel leaving the channel and to the shoaling (the reducing depth) not being detected until sixteen seconds before contact. It is extremely likely that the late detection, in turn, contributed to the grounding.

The chart plotter display very likely froze during the channel transit and did not recover. It is extremely likely that this led to the decision to continue. The fault had been reported verbally at the change of master but was not entered in the defect book. It is virtually certain this occurred, and likely that it contributed to the plotter freezing during the transit. Behind this was the operator's change-of-master defect brief, which is verbal, with nothing to check that what is said gets written down. It is virtually certain this existed, and likely that it contributed. This was identified as a safety issue.

The master had no plan to fall back on when the plotter was lost. There was no night passage plan for the channel and no hold point (a set place and condition at which the vessel stops) if the primary means of fixing position is lost. It is virtually certain this existed, and very likely that it contributed to the transit continuing. This gap traced back to the winter timetable, which moved the last service into darkness without the route risk assessment being reopened. It is extremely likely this occurred, and likely that it contributed. Both were identified as safety issues.

Without the plotter, the master looked for the No. 4 beacon, but it is extremely likely the beacon could not be seen from the channel over the half mile before the vessel left it. It is very likely this contributed to the vessel leaving the channel. Two conditions made the beacon harder to see, and both were identified as safety issues. First, the beacon had been converted to solar power, which halved its nominal range, and no operator was told. It is virtually certain this occurred, and likely that it contributed. Second, clearing of the beacon's sightline had been deferred, unfunded, for three works rounds. It is virtually certain this occurred, and very likely that it contributed. Both conditions stemmed from a wider safety issue: aids to navigation on this waterway carry no availability standard and no duty to tell users anything. It is virtually certain this existed, and likely that it contributed.

The investigation also examined whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up. The floodlighting is aligned with the beacon, and the masters' accounts of it were consistent and independent. However, the evidence could not establish either way whether it affected detection. The investigation assessed this as about as likely as not, and no conclusion is drawn. The question may not be answerable while the mangroves remain in place. Whatever the answer, the channel warrants a lighting assessment by the Authority.

Two other explanations were ruled out. It is very unlikely that the water level was below prediction, and very unlikely that a steering or propulsion failure caused the vessel to sheer off track.

What we found

Contributing safety factors

- The winter timetable very likely moved the last service into darkness without the route risk assessment being reopened, and this likely contributed to the grounding. (Safety issue)
- There was no night passage plan for the channel, and no hold point if the primary means of fixing position was lost; this was virtually certain, and it very likely contributed. (Safety issue)
- It is virtually certain that the transit continued at eight knots after the plotter froze, rather than the vessel stopping in the channel, and very likely that this contributed to the grounding.
- Shoaling was not detected until sixteen seconds before contact, which is virtually certain, and this very likely contributed.
- The change-of-master defect brief is verbal, with nothing to check that what is said gets written down; this is virtually certain, and it likely contributed. (Safety issue)
- With virtual certainty, the plotter fault was passed on verbally at the change of master and not entered in the defect book, and this likely contributed.
- The chart plotter display very likely froze during the channel transit and did not recover, and this very likely contributed to the grounding.
- Aids to navigation on this waterway carry no availability standard and no duty to tell users anything; this is virtually certain, and it likely contributed. (Safety issue)
- It is virtually certain that the No. 4 beacon was converted to solar power, halving its nominal range, with no operator told, and likely that this contributed. (Safety issue)
- The No. 4 beacon was very likely not visible from the channel over the half mile before the vessel left it, and this very likely contributed.
- With virtual certainty, the vessel left the dredged channel and tracked onto Boolarong Bank, and it is virtually certain that this contributed to the grounding.
- Clearing of the beacon's sightline was deferred, unfunded, for three works rounds, which is virtually certain, and this very likely contributed. (Safety issue)

Other factors that increased risk

- It is virtually certain that the forward saloon lifejacket locker cannot be reached from most of the saloon it serves. This did not influence the occurrence, but it very likely increases risk to passengers. (Safety issue)
- The master was, with virtual certainty, on day nine of twelve, on a split duty across a thirteen hour forty spread. This did not influence the occurrence, but it likely increases risk on the route's winter timetable.

Other findings

- The investigation examined whether a water level below prediction reduced the depth over Boolarong Bank, and ruled it out as very unlikely.
- A steering or propulsion failure causing the vessel to sheer off track was examined and ruled out as very unlikely.
- It is virtually certain that the engines were stopped and no attempt was made to refloat on a falling tide, and very likely that this reduced risk.

What is being done

Aids to navigation carry no availability standard and no duty to inform users. Before this report, the investigation raised its findings on the No. 4 beacon with the Port Authority and the operator. No action has been recorded in response to the scope of the Waterways Aids to Navigation Code WAN-2, section 4, as it applies to the Weerona Channel. [NOT ESTABLISHED: any action proposed or taken on the code's scope, and by whom]

Beacon converted to solar with no operator told. This was raised with the Port Authority and the operator. No action has been recorded. This includes any local notice to mariners or any correction to the operator's passage plan, which still showed the beacon at 6 M. [NOT ESTABLISHED: action taken and current status]

Sightline clearing deferred for three works rounds. This was raised with the Port Authority. No action has been recorded on the mangrove regrowth on Weerona Islet or on work order WO-176. [NOT ESTABLISHED: action taken and current status]

Winter timetable moved the last service into darkness without reopening the route risk assessment. On 08/07/2026 the General Manager Operations suspended the evening Muntari services. No review of RRA-07 has been recorded. [NOT ESTABLISHED: whether RRA-07 is being reviewed, and the conditions for resuming evening services]

No night passage plan and no hold point if the plotter is lost. The evening services remain suspended. No revision of OPS-3 or appendix A has been recorded. [NOT ESTABLISHED: action taken and current status]

Change-of-master defect brief is verbal and unchecked. No action has been recorded on OPS-9. No action has been recorded on applying the manufacturer's firmware fix issued in 11/2024. [NOT ESTABLISHED: action taken and current status]

Forward saloon lifejacket locker not reachable from most of the saloon. The same counter is fitted to the operator's two other 24 m vessels. The fleet-wide question is with the operator and the state regulator, and their answers are pending. [NOT ESTABLISHED: action taken on MV Tarrakin Star and the other two vessels, and on the minor modification route]

Distribution

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